MODULE 7

Above, Below, and Beyond

"Man cannot discover new oceans unless he has the courage to lose sight of the shore."

— ANDRÉ GIDE

ESSENTIAL QUESTION

What role does curiosity play in exploration?

MAIN SELECTIONS

1	**Into the Unknown: Above and Below**	· Stewart Ross (illus. Stephen Biesty)	INFORMATIONAL TEXT
2	**Great Discoveries and Amazing Adventures**	· Claire Llewellyn	INFORMATIONAL TEXT
3	**SpaceShipOne**	· Matthew Stinemetze (as told to Naomi Wallace)	AUTOBIOGRAPHY
4	**The Mighty Mars Rovers**	· Elizabeth Rusch	NARRATIVE NONFICTION

CONTENTS

A · Vocabulary (29) ..... p. 2

B · Reading Comprehension (32) ..... p. 6

C · IB Critical Thinking (5) ..... p. 10

D · MAP Practice (20) ..... p. 13

A Worksheet A — Vocabulary

Module 7 · Above, Below, and Beyond · Grade 5 · 29 Items

CCSS L.5.4 · L.5.5 · L.5.6

NAME _____ DATE _____

Directions. Read each question carefully. Watch out for tricky choices. Fill in the circle next to the BEST answer.

BIG IDEA WORDS

Q1 L.5.6

chronology

A **chronology** records —

- ◯ **A.** the price of a list of objects.
- ◯ **B.** the weather in a faraway place.
- ◯ **C.** the time and order of a series of events.
- ◯ **D.** the names of every country on Earth.

Q2 L.5.6

expedition

An **expedition** is —

- ◯ **A.** a trip that has a purpose, such as exploration or research.
- ◯ **B.** a short stroll taken close to home.
- ◯ **C.** a map drawn after a journey ends.
- ◯ **D.** a place where travelers stop to rest.

Q3 L.5.6

progress

To make **progress** is to —

- ◯ **A.** stop and turn back to the start.
- ◯ **B.** forget what you were doing.
- ◯ **C.** stay exactly where you began.
- ◯ **D.** improve or complete steps toward a goal.

Q4 L.5.5 TRAP

incredible

If something is **incredible**, it is —

- ◯ **A.** dull and easy to forget.
- ◯ **B.** so amazing that it is hard to believe.
- ◯ **C.** ordinary and expected.
- ◯ **D.** broken into many pieces.

CRITICAL VOCABULARY · INTO THE UNKNOWN: ABOVE AND BELOW

Q5 L.5.5 TRAP

vast

Something that is **vast** is —

- ◯ **A.** very large.
- ◯ **B.** tiny and hard to see.
- ◯ **C.** wet and cold.
- ◯ **D.** bright and noisy.

Q6 L.5.4

mariners

Mariners are —

- ◯ **A.** soldiers who guard a fort on land.
- ◯ **B.** scientists who study the weather.
- ◯ **C.** people, such as sailors, who navigate ships.
- ◯ **D.** pilots who fly high-altitude balloons.

Q7 L.5.4

cosmic

Something that is **cosmic** is —

- ◯ **A.** found deep inside the ground.
- ◯ **B.** beyond Earth and its atmosphere.
- ◯ **C.** made only of water.
- ◯ **D.** too small to measure.

Q8 L.5.5 TRAP

ascend

To **ascend** is to —

- ◯ **A.** move sideways.
- ◯ **B.** stand perfectly still.
- ◯ **C.** sink down to the bottom.
- ◯ **D.** go up.

Q9 L.5.4

principle

A scientific **principle** is —

- ◯ **A.** a rule that explains how something in the natural world works.
- ◯ **B.** a single experiment done one time.
- ◯ **C.** the leader of a school.
- ◯ **D.** a guess that has not been tested.

Q10 L.5.5 TRAP

forged

A metal object that was **forged** was —

- ◯ **A.** left to rust in the rain.
- ◯ **B.** made out of soft paper.
- ◯ **C.** heated in a special furnace and hammered into shape.
- ◯ **D.** frozen into a block of ice.

CRITICAL VOCABULARY · GREAT DISCOVERIES AND AMAZING ADVENTURES

Q11 L.5.4

ransacked

If someone **ransacked** a place, that person —

- ◯ **A.** cleaned it until it shone.
- ◯ **B.** damaged it while looking for something.
- ◯ **C.** painted it a bright new color.
- ◯ **D.** built it carefully from stone.

Q12 L.5.4

outskirts

The **outskirts** of a city or town are —

- **A.** its tallest buildings downtown.
- **B.** its main shopping street.
- **C.** the people who live there.
- **D.** its outer edges, farthest from the center.

Q13 L.5.5 TRAP

precious

Something that is **precious** —

- **A.** has value and should be treated with care.
- **B.** is worthless and easily thrown away.
- **C.** is common and found everywhere.
- **D.** is too heavy to lift.

Q14 L.5.4

authorities

Authorities are —

- **A.** tourists visiting a new country.
- **B.** books kept in a library.
- **C.** the people in charge, who have the power to make decisions or give orders.
- **D.** workers who paint old buildings.

Q15 L.5.5 TRAP

ruthless

Someone who is **ruthless** —

- **A.** is gentle and kind to everyone.
- **B.** shows no compassion and is very cruel.
- **C.** is shy and quiet around others.
- **D.** is helpful and generous to all.

CRITICAL VOCABULARY · SPACESHIPONE

Q16 L.5.4

ignited

Something that **ignited** —

- ◯ **A.** caught fire or exploded.
- ◯ **B.** turned into cold water.
- ◯ **C.** floated quietly in the air.
- ◯ **D.** melted slowly in the sun.

Q17 L.5.4

hybrid

Something can be described as **hybrid** if it is —

- ◯ **A.** made of only one material.
- ◯ **B.** too old to be used.
- ◯ **C.** painted a single color.
- ◯ **D.** made up of two or more different things.

Q18 L.5.4

synthetic

A **synthetic** item is —

- ◯ **A.** grown in a garden.
- ◯ **B.** found in the wild.
- ◯ **C.** artificial rather than natural.
- ◯ **D.** made entirely of stone.

CRITICAL VOCABULARY · SPACESHIPONE (CONTINUED)

Q19 L.5.4

exhaust

In an engine, **exhaust** is —

- ◯ **A.** the fuel that is added before takeoff.
- ◯ **B.** the gas that comes out as a waste product.
- ◯ **C.** the seat where the pilot sits.
- ◯ **D.** the metal frame of the ship.

Q20 L.5.4

physics

Physics is the scientific study of —

- ◯ **A.** matter and energy.
- ◯ **B.** ancient languages.
- ◯ **C.** plants and gardens.
- ◯ **D.** music and painting.

Q21 L.5.4

institution

An **institution** is —

- ◯ **A.** a single worker who acts alone.
- ◯ **B.** a rule that members must obey.
- ◯ **C.** a building left empty and unused.
- ◯ **D.** an organization that has a particular focus or goal.

CRITICAL VOCABULARY · THE MIGHTY MARS ROVERS

Q22 L.5.4

deploy

If you **deploy** something, you —

- ◯ **A.** take it apart and throw it away.
- ◯ **B.** move it into position so it can be used.
- ◯ **C.** hide it where no one can find it.
- ◯ **D.** leave it locked in a box.

Q23 L.5.5 TRAP

transition

In a **transition**, —

- ◯ **A.** everything stays exactly the same.
- ◯ **B.** two objects are weighed.
- ◯ **C.** one thing changes to something else.
- ◯ **D.** a price is written down.

Q24 L.5.4

expanse

An **expanse** is —

- ◯ **A.** a very large area of land, sea, or sky.
- ◯ **B.** a narrow crack in a rock.
- ◯ **C.** a small box for tools.
- ◯ **D.** a short message sent quickly.

Q25 L.5.5 TRAP

resembled

If two things **resembled** each other, they —

- ◯ **A.** were kept far apart.
- ◯ **B.** were made of metal.
- ◯ **C.** moved in opposite directions.
- ◯ **D.** looked like each other.

CRITICAL VOCABULARY · THE MIGHTY MARS ROVERS (CONTINUED)

Q26 L.5.4

terrain

The **terrain** of an area is —

- ◯ **A.** the weather in the sky above it.
- ◯ **B.** what the surface of the land looks like.
- ◯ **C.** the people who live on it.
- ◯ **D.** the name given to the area.

Q27 L.5.4

international

Something that is **international** is —

- ◯ **A.** kept secret inside one home.
- ◯ **B.** built only of glass and steel.
- ◯ **C.** shared or worked on by multiple countries.
- ◯ **D.** finished in a single day.

Q28 L.5.4

transmitted

When an electronic message is **transmitted**, it is —

- ◯ **A.** sent from one place to another.
- ◯ **B.** erased before anyone reads it.
- ◯ **C.** printed onto thick paper.
- ◯ **D.** folded and stored away.

Q29 L.5.5 TRAP

ailing

When something is **ailing**, it is —

- ◯ **A.** healthy and working perfectly.
- ◯ **B.** brand new and powerful.
- ◯ **C.** moving faster than before.
- ◯ **D.** not doing well or getting weak.

Worksheet B — Reading Comprehension

Module 7 · Above, Below, and Beyond · Grade 5 · 32 Items

CCSS RI.5.1 · RI.5.2 · RI.5.3 · RI.5.5 · RI.5.6 · RI.5.8

NAME _____ DATE _____

Directions. Use what you remember from each selection to answer the questions. Fill in the circle next to the BEST answer.

Selection 1 · Into the Unknown: Above and Below

INFORMATIONAL TEXT · STEWART ROSS (ILLUS. STEPHEN BIESTY) · 8 QUESTIONS

Q1 RI.5.2

What is "Into the Unknown: Above and Below" mostly about?

- ◯ **A.** A class that builds a model rocket for a science fair.
- ◯ **B.** Two friends who sail around the world for fun.
- ◯ **C.** The Piccard family's record journeys to the heights of the sky and the depths of the sea.
- ◯ **D.** A team of robots sent to explore the planet Mars.

Q2 RI.5.1

According to the text, why were Auguste and Jacques Piccard unlike many earlier explorers?

- ◯ **A.** They were scientists, not soldiers or hardy mariners.
- ◯ **B.** They explored only on foot.
- ◯ **C.** They never left their home country.
- ◯ **D.** They refused to use any machines.

Q3 RI.5.3

According to the text, why did Auguste Piccard want to travel high into the stratosphere?

- ◯ **A.** To take photographs of his hometown.
- ◯ **B.** To set a record for the fun of it alone.
- ◯ **C.** To find a faster path across the ocean.
- ◯ **D.** To observe the cosmic rays he had grown interested in as a professor of physics.

Q4 RI.5.3

According to the text, how did Auguste Piccard's sealed cabin keep him safe high in the stratosphere?

- **A.** It was filled with ocean water.
- **B.** Once sealed, it became airtight so the pressure inside stayed steady, with a supply of oxygen to breathe.
- **C.** It floated on top of a balloon made of steel.
- **D.** It opened its hatches to let in fresh mountain air.

Q5 RI.5.3

According to the text, what record did Auguste Piccard set on his 1931 balloon flight?

- **A.** He sailed farther than any ship before him.
- **B.** He stayed underwater longer than anyone.
- **C.** He crossed an entire ocean by balloon.
- **D.** He and his assistant reached a world-record height in the stratosphere.

Q6 RI.5.8

What evidence in the text supports the idea that exploring the deep ocean was extremely dangerous?

- **A.** In the deepest seas, the water can press hard enough to crush an ordinary submarine.
- **B.** The ocean is always warm and calm near the bottom.
- **C.** Sailors had already mapped every part of the sea.
- **D.** The deep ocean has plenty of light for divers.

Q7 RI.5.5

How is this informational text mainly organized, and how do its visuals help readers?

- **A.** As a rhyming poem with no pictures.
- **B.** As a set of cooking directions.
- **C.** In time order, with detailed cutaway diagrams that show how the balloon and the deep-sea cabin worked.
- **D.** As a play with characters speaking lines.

Q8 RI.5.2

Which sentence BEST states a central idea of this selection?

- **A.** A balloon filled with light hydrogen gas can rise into the cold stratosphere.
- **B.** Scientific curiosity and courage drove the Piccards to explore the most extreme heights and depths of our world.
- **C.** The Trieste was towed out to sea before its record-setting dive.
- **D.** A sealed cabin must stay airtight to keep its crew safe at extreme heights.

Selection 2 · Great Discoveries and Amazing Adventures

INFORMATIONAL TEXT · CLAIRE LLEWELLYN · 8 QUESTIONS

Q9 RI.5.2

What is "Great Discoveries and Amazing Adventures" mostly about?

- ◯ **A.** A family that travels to the edge of space.
- ◯ **B.** A pilot who builds his own airplane.
- ◯ **C.** A robot that drives across another planet.
- ◯ **D.** Hidden marvels and lost treasures that explorers and archaeologists uncovered.

Q10 RI.5.3

According to the text, what did Hiram Bingham bring to the world's attention in 1911?

- ◯ **A.** A new way to build steel bridges.
- ◯ **B.** The "lost" Inca city of Machu Picchu.
- ◯ **C.** A cave full of ancient paintings in France.
- ◯ **D.** A buried army of clay soldiers in China.

Q11 RI.5.3

According to the text, how were the Lascaux cave paintings first found?

- ◯ **A.** Boys rescuing their dog from a hole found that it led to a cavern.
- ◯ **B.** A scientist dug them up on purpose.
- ◯ **C.** They were shown on an old map.
- ◯ **D.** A farmer plowed them up in a field.

Q12 RI.5.1

According to the text, who uncovered the Terra-Cotta Army built for Emperor Qin Shi Huangdi?

- ◯ **A.** The explorer Hiram Bingham.
- ◯ **B.** Four French schoolboys.
- ◯ **C.** The archaeologist Yuan Zhongyi.
- ◯ **D.** A team of mariners from Spain.

Q13 RI.5.3

According to the text, why was the discovery of the Lascaux paintings later a problem to protect?

- ◯ **A.** Crowds of visitors dampened the cave walls and damaged the precious paintings.
- ◯ **B.** The paintings were too small to see.
- ◯ **C.** No one was interested in them.
- ◯ **D.** The cave was located on a high mountain.

Q14 RI.5.6

Why does the author include quotations and details about how each discovery was made?

- ◯ **A.** To persuade readers to travel to Peru on vacation.
- ◯ **B.** To teach readers how to dig up ancient ruins themselves.
- ◯ **C.** To warn readers that exploring old sites is too dangerous.
- ◯ **D.** To support the central ideas with relevant facts about real discoveries.

Q15 RI.5.5

How does the author mainly organize this informational text?

- ◯ **A.** As one long rhyming poem.
- ◯ **B.** With headings and subheadings, each telling the story of a different discovery, plus photos and captions.
- ◯ **C.** As a set of math problems.
- ◯ **D.** As a play with scenes and actors.

Q16 RI.5.2

Which sentence BEST states a central idea of this selection?

- ◯ **A.** Hiram Bingham climbed steep mountain paths to reach Machu Picchu.
- ◯ **B.** Crowds of visitors once damaged the precious Lascaux cave paintings.
- ◯ **C.** Curiosity and persistence lead explorers to uncover hidden treasures that connect us with ancient peoples.
- ◯ **D.** The Terra-Cotta Army was buried to guard one emperor's tomb.

Selection 3 · SpaceShipOne

AUTOBIOGRAPHY · MATTHEW STINEMETZE (AS TOLD TO NAOMI WALLACE) · 8 QUESTIONS

Q17 RI.5.2

What is "SpaceShipOne" mostly about?

- ◯ **A.** A family balloon trip into the stratosphere.
- ◯ **B.** A team that built and flew a private spacecraft to the edge of space.
- ◯ **C.** Explorers searching for treasure in old caves.
- ◯ **D.** Two robots exploring the planet Mars.

Q18 RI.5.6

How can readers tell that this selection is an autobiography?

- ◯ **A.** It rhymes from beginning to end.
- ◯ **B.** It is written entirely as a list of facts with no narrator.
- ◯ **C.** It is told by someone who never saw the events.
- ◯ **D.** The narrator tells his own true story using first-person words like *I* and *we*.

Q19 RI.5.3

According to the text, how did SpaceShipOne reach the sky before its rocket fired?

- **A.** It attached to the belly of a mother ship, the White Knight, which carried it up.
- **B.** It was launched straight up from the ground.
- **C.** It was carried up by a giant balloon.
- **D.** It was dropped from a tall mountain.

Q20 RI.5.3

According to the text, what happened after SpaceShipOne was released from the mother ship?

- **A.** The ship landed back on the runway at once.
- **B.** The crew opened the hatch to let in air.
- **C.** The pilot ignited the rocket engine and shot almost straight up toward space.
- **D.** The ship floated slowly down to the sea.

Q21 RI.5.1

Which detail BEST shows that SpaceShipOne used an unusual kind of rocket?

- **A.** It used a hybrid rocket fueled by a mixture of two substances.
- **B.** It was painted white and shiny.
- **C.** It had a number painted on its side.
- **D.** It carried a single pilot.

Q22 RI.5.6

Why does the narrator share his excited feelings, such as "We are at the beginning of something huge"?

- **A.** To give plain technical facts about how the rocket worked.
- **B.** To complain that the project took too much time and effort.
- **C.** To share his own point of view and show why he believed the flight was historic.
- **D.** To warn readers that space travel is too dangerous to try.

Q23 RI.5.3

According to the text, which school subjects helped the narrator earn his job on the project?

- **A.** Only spelling and handwriting.
- **B.** Cooking and sewing.
- **C.** Music and dance only.
- **D.** Math, physics, science, and art.

Q24 RI.5.2

Which sentence BEST states a central idea of "SpaceShipOne"?

- **A.** The White Knight carried SpaceShipOne up before its rocket fired.
- **B.** Curiosity, teamwork, and hard work made private space exploration possible and historic.
- **C.** SpaceShipOne could take off from almost any airport in the world.
- **D.** The narrator's school subjects included math, physics, science, and art.

Selection 4 · The Mighty Mars Rovers

NARRATIVE NONFICTION · ELIZABETH RUSCH · 8 QUESTIONS

Q25 RI.5.2

What is "The Mighty Mars Rovers" mostly about?

- **A.** A scientist and his team sending twin robots, Spirit and Opportunity, to explore Mars.
- **B.** A balloon flight into the stratosphere.
- **C.** A pilot flying a spacecraft to the edge of space.
- **D.** Explorers digging up an ancient city.

Q26 RI.5.3

According to the text, what did scientist Steve Squyres imagine for many years?

- **A.** A balloon that could reach the stratosphere.
- **B.** A submarine that could reach the ocean floor.
- **C.** A museum filled with rockets.
- **D.** A robotic geologist that could explore Mars and send data back to Earth.

Q27 RI.5.3

According to the text, why was the rovers' landing so risky?

- **A.** The rovers were too heavy to move at all.
- **B.** If the lander did not pass through the atmosphere just right, it would burn up or be smashed.
- **C.** The team had never tested any equipment.
- **D.** The rovers had to be driven by a person on board.

Q28 RI.5.3

According to the text, how did the team feel right after the first signal was lost following the landing?

- **A.** They were bored and went home.
- **B.** They were certain nothing had gone wrong.
- **C.** Their cheering stopped and they were worried when the signal suddenly cut off.
- **D.** They decided to end the mission for good.

Q29 RI.5.8

What evidence does the text give that water once existed on Mars?

- **A.** Evidence pointed again and again to a flowing, salty body of water that once existed on Mars.
- **B.** The rovers found a flowing river still running on the surface.
- **C.** The team saw rain falling from Martian clouds.
- **D.** The rovers brought back a bottle of water.

Q30 RI.5.5

How is this narrative nonfiction mainly organized?

- **A.** As a list of unrelated facts in any order.
- **B.** As a rhyming poem about space.
- **C.** As a true story told in time order, sharing real people's feelings as the mission unfolds.
- **D.** As a set of recipes.

Q31 RI.5.6

Why does the author describe the team's hugs, cheers, tears, and worry during the mission?

- **A.** To show that the scientists did not really care if the rovers failed.
- **B.** To give dry technical facts about how the landing worked.
- **C.** To prove that the team wasted time celebrating instead of working.
- **D.** To help readers feel the suspense and understand how much the people cared about the mission.

Q32 RI.5.2

Which sentence BEST states a central idea of "The Mighty Mars Rovers"?

- **A.** The twin rovers were named Spirit and Opportunity.
- **B.** Patience, persistence, and curiosity drive the scientific exploration of Mars.
- **C.** Opportunity found salts that hinted water once existed on Mars.
- **D.** Each rover bounced inside airbags as it landed on Mars.

Worksheet C — IB Critical Thinking

Module 7 · Above, Below, and Beyond · Grade 5

IB PYP Key Concepts · DOK 2 – 4

NAME _____ DATE _____

Directions. Use what you learned from each selection to think more deeply about how curiosity drives exploration. Each question helps you grow as an IB learner.

Q1 Curiosity DOK 3

In **"Into the Unknown: Above and Below,"** the Piccards explored places no one had reached. Show a QUESTION that made them curious and the DANGER they faced, then explain how curiosity led to discovery.

CURIOSITY & DANGER · THE PICCARDS

A QUESTION THAT MADE THEM CURIOUS	A DANGER THEY FACED

How did curiosity lead the Piccards to explore the heights and depths of our world? Use the word *curiosity* in your answer.

Q2 Discovery DOK 3

In **"Great Discoveries and Amazing Adventures,"** explorers uncovered hidden treasures. Show the SEARCH and the DISCOVERY, then explain why these discoveries matter today.

SEARCH AND DISCOVERY · HIDDEN MARVELS

A SEARCH OR QUESTION	→	WHAT WAS DISCOVERED

Why do discoveries like Machu Picchu or the Terra-Cotta Army matter to people today? Use one detail from the text.

Q3 Perseverance DOK 3

In **"SpaceShipOne,"** a team worked hard to fly a private spacecraft. Show the STEPS, then explain how teamwork and effort made the flight possible.

HOW THE FLIGHT HAPPENED · SPACESHIPONE

THE PLAN	THE WORK	THE RESULT
how did the ship reach the sky?	*what did the team do?*	*what did the flight achieve?*

How did teamwork and hard work make this space flight possible? Use one detail from the selection.

Q4 Courage DOK 3

In **"The Mighty Mars Rovers,"** a team faced great risks to explore Mars. Show a RISK they faced and how they RESPONDED, then explain why patience and courage mattered.

RISK AND RESPONSE · MARS ROVERS

A RISK OR SETBACK THEY FACED	HOW THE TEAM RESPONDED

Why did patience and courage matter for the team that explored Mars? Explain using one detail.

Q5 Inquiry DOK 4

Essential Question: What role does curiosity play in exploration? Use ideas from TWO of the selections in this module to answer.

TWO-SELECTION SYNTHESIS

SELECTION 1 · TITLE

What did curiosity lead the explorers to do?

SELECTION 2 · TITLE

What did curiosity lead the explorers to do?

MY BIG IDEA

Worksheet D — MAP Practice

Module 7 · Above, Below, and Beyond · Grade 5

MAP RIT 205 → 225 · 4 Mini-Passages × 5 Questions

NAME _____ DATE _____

Directions. Read each short passage carefully. Then answer the 5 questions that follow. Each passage is NEW — not from the selections in this workbook.

Passage 1 · The Cave Behind the Waterfall

Maya had hiked the trail behind her grandmother's house a hundred times, but she had never gone past the small waterfall at its end. One cool morning she noticed something she had missed before. Behind the curtain of falling water, there seemed to be a dark gap in the rock. Her heart beat faster. What if it was a cave? Maya did not run in right away. She remembered her grandmother's rule: never explore alone. So she went home, found a flashlight, and asked her older brother to come along. Together they stepped carefully behind the water. The gap really was the mouth of a small cave. Inside, the walls were cool and damp, and their lights showed strange shapes in the stone. There were no monsters, only quiet rooms shaped by water over many years. Maya did not touch anything. She knew a place this old should be treated with care. On the way home, she could not stop talking. She had walked past the waterfall a hundred times, yet only her curiosity, and a careful plan, had finally led her to look closer and find something new.

189 WORDS · LEXILE 620–700L · RIT 205–211

Q1 RIT 205 · Lexile 620L · Key Detail

What did Maya notice one cool morning that she had missed before?

- ◯ **A.** A boat tied up at the river.
- ◯ **B.** A dark gap in the rock behind the waterfall.
- ◯ **C.** A path leading up a mountain.
- ◯ **D.** A nest full of baby birds.

Q2 RIT 207 · Lexile 640L · Key Detail

Why did Maya not run into the cave right away?

- ◯ **A.** She was not interested in it.
- ◯ **B.** The water was too cold to cross.
- ◯ **C.** She remembered her grandmother's rule never to explore alone.
- ◯ **D.** She had forgotten the way home.

Q3 RIT 208 · Lexile 660L · Cause & Effect

What did Maya do before going behind the water?

- ◯ **A.** She told no one and went in alone.
- ◯ **B.** She gave up and went back inside.
- ◯ **C.** She waited for the waterfall to stop.
- ◯ **D.** She went home, found a flashlight, and asked her brother to come along.

Q4 RIT 210 · Lexile 685L · Inference

Why did Maya choose not to touch anything inside the cave?

- ◯ **A.** She knew a place this old should be treated with care.
- ◯ **B.** The walls were too high to reach.
- ◯ **C.** She was in a hurry to leave.
- ◯ **D.** Her brother told her to run.

Q5 RIT 211 · Lexile 700L · Theme

What is the main message, or theme, of this passage?

- ◯ **A.** A walk you have taken many times never holds anything new.
- ◯ **B.** It is safest to explore an unknown place completely on your own.
- ◯ **C.** Curiosity, along with a careful plan, can help a person discover something new.
- ◯ **D.** Old caves should be touched so you can learn how they feel.

Passage 2 · How Deep-Sea Explorers See in the Dark

The deep ocean is one of the hardest places on Earth to explore. Sunlight cannot reach the bottom of the deepest seas, so the water there is completely dark and very cold. The weight of the water pressing down is so great that it would crush an ordinary submarine. For these reasons, people cannot simply swim down to study the deep sea. Instead, scientists use special machines. Some are called submersibles. A submersible is a small craft with thick, strong walls that can carry a crew safely into the deep. Its powerful lights cut through the darkness so the crew can see the seafloor. Cameras record everything the crew finds, and mechanical arms can pick up samples of rock or reach out toward sea creatures. Today, many of these craft do not even need people inside. A robot submersible can be guided from a ship on the surface, sending pictures up through a long cable. Thanks to these tools, explorers have discovered strange fish, towering underwater mountains, and animals that glow in the dark. Each dive reveals more of a hidden world that humans had never seen before, proving how much is still left to explore.

196 WORDS · LEXILE 740–840L · RIT 211–217

Q6 RIT 211 · Lexile 740L · Key Detail

According to the passage, why can people not simply swim down to study the deep sea?

- **A.** It is dark, very cold, and the water presses down hard enough to crush an ordinary submarine.
- **B.** The deep sea is too warm and bright.
- **C.** There is nothing interesting to see there.
- **D.** The water is too shallow for diving.

Q7 RIT 213 · Lexile 770L · Key Detail

According to the passage, what is a submersible?

- **A.** A kind of fish that glows in the dark.
- **B.** A large balloon used over the ocean.
- **C.** A camera attached to a diver's suit.
- **D.** A small craft with thick, strong walls that can carry a crew into the deep.

Q8 RIT 214 · Lexile 800L · Vocabulary

In the passage, the words **cut through the darkness** show that the lights —

- **A.** made the water colder.
- **B.** were bright enough to let the crew see in the dark.
- **C.** sliced the walls of the craft.
- **D.** turned off to save power.

Q9 RIT 215 · Lexile 820L · Author's Purpose

Why does the author describe robot submersibles guided from a ship?

- **A.** To warn readers that the deep sea is too risky to study at all.
- **B.** To prove that crewed submersibles are no longer used anywhere.
- **C.** To show how new tools let people explore the deep sea without going down themselves.
- **D.** To explain step by step how to build a submersible.

Q10 RIT 217 · Lexile 840L · Main Idea

What is the main idea of "How Deep-Sea Explorers See in the Dark"?

- **A.** The cameras on a submersible are its single most important part.
- **B.** The deep ocean is too cold and dark for anything to live there.
- **C.** Robot submersibles have completely replaced crewed ones today.
- **D.** Special machines help people explore and discover a hidden deep-sea world they could not reach before.

Passage 3 · The Night Sky Club

Theo had always believed that stargazing was boring. To him, the night sky was just a few dim dots scattered above the city. Then his cousin Ana invited him to join her Night Sky Club, which met far from the bright city lights. Theo almost said no, but he was a little curious, so he went. When they arrived at the dark field, Theo looked up and gasped. The sky was not empty at all. Thousands of stars spread across it like spilled sugar, brighter than he had ever imagined. Ana handed him a small telescope. "Try this," she said. "Find the moon." Theo pointed the telescope and held his breath. Suddenly the moon filled the lens, close enough to show its craters and shadows. He had seen the moon his whole life, yet he had never really looked at it. For the next hour, Ana showed him how to find planets and patterns of stars. Theo asked question after question, and the more he learned, the more he wanted to know. On the drive home, he was already planning his next visit. The sky he had once thought was boring had become the most amazing thing he had ever seen.

201 WORDS · LEXILE 760–860L · RIT 217–222

Q11 RIT 217 Lexile 760L Character

How did Theo feel about stargazing at the START of the passage?

- **A.** He thought it was the best hobby in the world.
- **B.** He believed it was boring.
- **C.** He was afraid of the dark.
- **D.** He had never seen the night sky.

Q12 RIT 218 Lexile 790L Key Detail

Why did Theo agree to go to the Night Sky Club?

- **A.** He was a little curious.
- **B.** His parents made him go.
- **C.** He wanted to use a new phone.
- **D.** The club met inside the city.

Q13 RIT 219 Lexile 820L Cause & Effect

What happened when Theo pointed the telescope at the moon?

- **A.** The telescope broke.
- **B.** Clouds covered the sky.
- **C.** He could not find the moon at all.
- **D.** The moon filled the lens, close enough to show its craters and shadows.

Q14 RIT 220 Lexile 840L Inference

What does it suggest that Theo "asked question after question" and "wanted to know" more?

- **A.** He was tired and ready to go home.
- **B.** He was annoyed with his cousin.
- **C.** His curiosity had been sparked, and he was eager to learn more.
- **D.** He still thought the sky was boring.

Q15 RIT 222 · Lexile 860L · Theme

What is the main message, or theme, of this passage?

- ◯ **A.** Being curious enough to look closer can turn something ordinary into something amazing.
- ◯ **B.** A hobby you decide is boring will always stay boring to you.
- ◯ **C.** You can only enjoy the night sky with an expensive telescope.
- ◯ **D.** The best way to see stars is from the middle of a bright city.

Passage 4 · Why We Send Robots to Explore

When people imagine exploring space or the deep sea, they often picture a brave human in a special suit. Yet many of the most important discoveries are made by machines. There are good reasons to send a robot where a person cannot easily go. The first reason is safety. Some places are far too dangerous for humans. The surface of another planet may have no air to breathe, and the bottom of the ocean has enough pressure to crush a submarine. A robot can travel to these places without putting a human life at risk. The second reason is time. A trip to a distant planet can take many months or even years. A robot does not need food, water, or sleep, so it can keep working long after a human crew would have to turn back. The third reason is cost. Sending a person safely into space requires huge amounts of fuel and equipment, while a robot can be much lighter and cheaper to launch. Of course, robots are not perfect. They cannot make every choice on their own, so scientists guide them from far away. Still, by sending machines into the unknown, people can explore worlds that would otherwise stay beyond their reach.

205 WORDS · LEXILE 860–960L · RIT 219–225

Q16 RIT 219 · Lexile 860L · Key Detail

According to the passage, what is the FIRST reason given for sending a robot instead of a person?

- ◯ **A.** Robots are more fun to watch.
- ◯ **B.** Robots can take better photographs.
- ◯ **C.** Safety — some places are far too dangerous for humans.
- ◯ **D.** Robots can talk to people.

Q17 RIT 220 · Lexile 890L · Key Detail

According to the passage, why is time a reason to use a robot?

- ◯ **A.** A robot can fly faster than light.
- ◯ **B.** A robot does not need food, water, or sleep, so it can keep working when a human crew would have to turn back.
- ◯ **C.** A robot finishes every trip in one day.
- ◯ **D.** A robot never stops to take pictures.

Q18 RIT 221 | Lexile 910L | Cause & Effect

According to the passage, why is a robot often cheaper to launch than a human crew?

- **A.** A robot is made of gold.
- **B.** A robot is painted a special color.
- **C.** A robot needs no fuel at all.
- **D.** A robot can be much lighter, while sending a person safely needs huge amounts of fuel and equipment.

Q19 RIT 223 | Lexile 935L | Author's Purpose

Why does the author admit that "robots are not perfect"?

- **A.** To give a fair, balanced view before concluding that robots still help us explore.
- **B.** To argue that robots are too flawed and should never be used.
- **C.** To prove that humans are always better explorers than robots.
- **D.** To explain exactly how scientists guide a robot from far away.

Q20 RIT 225 | Lexile 960L | Main Idea

What is the main idea of "Why We Send Robots to Explore"?

- **A.** Because robots can fail, people should always go exploring themselves.
- **B.** For reasons of safety, time, and cost, robots let people explore places that would otherwise stay out of reach.
- **C.** Saving money is the only reason scientists ever send a robot.
- **D.** A robot can make every decision on its own, with no human help.